

PANEER SELVAM V

[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

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Skills

- **Languages/Frameworks** - Javascript, Typescript, NodeJs, ReactJs, NextJs
- **Database/Tools** - Postgresql, MongoDB, Prisma ORM
- **Others** - DSA, Problem Solving, Design Patterns

Projects

Project 3

Ben10 Alien Explorer

- Developed a full-stack Ben 10 explorer using Next.js, integrating searchable alien directories, series filtering, detailed alien pages, and an interactive Omnitrix-inspired interface.
 - Engineered Assist10, a domain-specific RAG chat system using Gemini embeddings, Astra DB vector search, and streamed Gemini responses, retrieving the six most relevant knowledge chunks before generation.
 - Designed a relational data architecture with PostgreSQL and Prisma for alien records, authentication, favorites, chat history, and feedback, while using Cloudinary for image storage.
 - Implemented authenticated user workflows with Better Auth, relational favorite constraints, Zod input validation, and a per-user process-local rate limiter to handle protected API operations and invalid requests.
 - Optimized data fetching and UI behavior through concurrent alien API requests, React Query caching, reducer-based explorer state, dynamic imports, and selective RAG context retrieval.
- **Tech - NextJs, Typescript, MongoDB, Postgresql, FramerMotion, NodeJs, Langchain**

Project 2

Custom Promise Library

- Engineered a custom JavaScript Promise/A+ implementation to explore promise states, asynchronous execution, and interoperability with promise-based workflows.
 - Implemented a microtask queue demonstration using `queueMicrotask` to illustrate asynchronous state updates and their consistency compared with `setTimeout`.
 - Designed a state-driven ordering visualizer that orchestrates pizza order transitions from placed → preparing → baking → delivery, synchronizing UI updates with the Promise lifecycle.
 - Separated UI logic (`ph.js`) from order processing (`orderService.js`) to improve code organization, maintainability, and debugging.
 - Implemented centralized error handling with order cancellation logic to demonstrate graceful handling of rejected asynchronous operations and unhappy paths.
- **Tech - JavaScript, Promises**

project 1

Code Editor

- Developed a browser-based code editor using Vanilla JavaScript and Ace Editor, enabling real-time HTML, CSS, and JavaScript editing with live preview through an iframe `srcdoc` document.
 - Engineered a sandboxed iframe execution workflow with lightweight HTML, CSS, and JavaScript validation to provide pre-execution feedback and structured browser-based code rendering.
 - Implemented a client-side project persistence system using `localStorage`, JSON serialization, and File/Blob APIs, supporting project restoration, import, and export with backward-compatible data normalization.
 - Designed modular browser services for storage, file handling, and logging, alongside keyboard shortcuts and accessible editor tab navigation to organize application behavior and improve usability.
- **Tech - HTML, CSS, JavaScript**

Education

Hindusthan college of Arts and Science

Bachelor of Science - Computer Science, CGPA – 7.4

2022 – 2025